



# SOFTWARE REQUIREMENTS SPECIFICATION

## *Food Management System*

### SRS DOCUMENT

#### USER CLASSES

Customer

Administrator

Delivery Staff

# SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

## FOOD MANAGEMENT SYSTEM

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### 1. INTRODUCTION

#### 1.1 Purpose

The purpose of the Food Management System is to develop a computerized system for managing food-related activities in an efficient and organized manner. The system provides a platform for managing food items, menus, customers, orders, payments, delivery, and feedback.

The system reduces manual work and helps administrators manage food and order information accurately. Customers can easily view the available food items, place orders, make payments, and track their orders.

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#### 1.2 Scope of the Project

The Food Management System is designed to manage the complete food ordering and management process.

The system provides the following major facilities:

- User registration and login
- Food and menu management
- Food search and selection
- Cart management
- Food ordering

- Order management
- Payment management
- Delivery status management
- Customer feedback and rating
- Admin management and reports

The system can be used by restaurants, canteens, food outlets, or similar food-service organizations.

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### 1.3 Definitions and Acronyms

Term| Definition

SRS| Software Requirements Specification

Admin| Person responsible for managing the system

Customer| User who views and orders food

UI| User Interface

DB| Database

UML| Unified Modeling Language

ER| Entity Relationship

Order ID| Unique identification number for an order

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### 1.3 References

The following concepts and technologies are considered while designing the system:

- Software Engineering principles

- UML modelling
- Database Management System concepts

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Web application development concepts

MySQL database concepts

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## 2. OVERALL DESCRIPTION

### 2.1 Product Perspective

The Food Management System is a software application that provides a centralized platform for managing food items and customer orders.

The system consists of different modules such as:

- Customer Management
- Admin Management
- Food/Menu Management
- Cart Management
- Order Management
- Payment Management
- Delivery Management
- Feedback Management

The system stores the required information in a database so that authorized users can access and manage the information efficiently.

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### 2.2 Product Functions

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The major functions of the system are:

1. User registration and login
2. Admin login
3. Adding and managing food items
4. Viewing food categories and menu
5. Searching for food items
6. Adding food items to cart
7. Placing orders
8. Calculating total order amount
9. Managing payment information
10. Updating order status
11. Managing delivery information
12. Providing feedback and ratings
13. Generating basic reports

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## 2.3 User Classes and Characteristics

### 2.3.1 Customer

The customer can:

- Register an account
- Login to the system

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- View the food menu
- Search for food
- Add food items to cart
- Place an order
- Make payment
- View order status
- View order history
- Give feedback and ratings

### 2.3.2 Administrator

The administrator manages the entire system.

The admin can:

- Login securely
- Add new food items
- Update food information
- Delete food items
- Manage food categories
- View customer details
- View and manage orders
- Update order status
- View payment information
- View feedback
- Generate reports

### 2.3.3 Delivery Staff

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The delivery staff can:

- Login to the system

View assigned orders

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View delivery details

Update delivery status

Mark orders as delivered

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### 2.3 Operating Environment

The system can operate in the following environment:

Hardware:

- Computer/Laptop/Smartphone
- Minimum 4 GB RAM
- Minimum 10 GB available storage
- Internet connection

Software:

- Windows/Linux/Android
  - Web browser such as Google Chrome, Microsoft Edge or Firefox
  - Backend server
  - MySQL database
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## 2.5 Design and Implementation Constraints

The following constraints are considered:

The system requires a database for storing information.

Users must provide valid login credentials.

Only authorized administrators can modify food and order information.

Internet connectivity may be required for online operations.

- Payment processing depends on the selected payment method or payment service.
- The system should protect customer information from unauthorized access.

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## 2.4 Assumptions and Dependencies

The system is developed based on the following assumptions:

- Customers have access to a valid account.
- Food information entered by the admin is accurate.
- The database server is available when the system is being used.
- Customers provide correct delivery information.
- The payment service is available during online payment.
- Delivery staff update the order status correctly.

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### 3. SYSTEM REQUIREMENTS

#### 3.1 Functional Requirements

##### FR1 – User Registration

The system shall allow new customers to create an account by providing:

- Name
- Email
- Phone number
- Address
- Password

The system shall validate the entered information before creating the account.

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##### FR2 – User Login

The system shall allow registered users to login using their registered email/username and password.

The system shall reject invalid login credentials.

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### FR3 – Admin Login

The system shall provide a secure login facility for administrators.

Only authorized administrators shall be allowed to access admin functions.

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### FR4 – Food Management

The admin shall be able to:

- Add food items
- Edit food details
- Delete food items

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Update food price

Update food availability

- Add food images

- Assign food to categories

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#### FR5 – Menu Viewing

Customers shall be able to view the available food menu.

The menu shall display:

- Food name

- Food category

- Description

- Price

- Availability

- Image

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#### FR6 – Food Search

The system shall allow customers to search for food items using the food name or category.

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#### FR7 – Cart Management

Customers shall be able to:

- Add food items to the cart
  - Remove food items
  - Increase or decrease quantity
  - View the total amount
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#### FR8 – Order Placement

The system shall allow customers to place an order after selecting the required food items.

The system shall generate a unique Order ID for every successful order.

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#### FR9 – Order Management

The admin shall be able to view and manage customer orders.

The order status may include:

- Pending
- Confirmed
- Preparing

- Ready
  - Out for Delivery
  - Delivered
  - Cancelled
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#### FR10 – Payment Management

The system shall calculate the total order amount and record payment details.

Payment information may include:

- Payment ID
  - Order ID
  - Amount
  - Payment method
  - Payment status
  - Payment date
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#### FR11 – Delivery Management

The system shall allow delivery staff to view assigned orders and update delivery status.

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#### FR12 – Order History

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Customers shall be able to view their previous orders and their details.

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#### FR13 – Feedback and Rating

Customers shall be able to provide feedback and ratings after receiving an order.

The admin shall be able to view customer feedback.

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#### FR14 – Report Generation

The admin shall be able to view basic reports such as:

- Total orders
  - Completed orders
  - Cancelled orders
  - Available food items
  - Customer information
  - Sales/payment information
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## 4. NON-FUNCTIONAL REQUIREMENTS

### 4.1 Performance

- The system should respond to user requests within a reasonable time.
- The system should support multiple users.

- Database operations should be performed efficiently.
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#### 4.2 Security

- User passwords must be securely stored.  
Only authorized users should access protected functions.
  - Admin functions must be restricted to administrators.
  - Customer information must be protected from unauthorized access.
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#### 4.3 Usability

- The system should have a simple and user-friendly interface.
  - Navigation should be easy to understand.
  - Food items should be displayed clearly.
  - Error messages should be understandable.
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#### 4.3 Reliability

- The system should work consistently without frequent failures.
  - Order information should not be lost.
  - Database information should be maintained accurately.
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#### 4.5 Availability

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- The system should be available whenever the service is being provided.
  - The database should be accessible during system operation.
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#### 4.4 Maintainability

- The system should be easy to maintain.
  - New food items and categories should be easily added.
  - Future features should be added without major changes to the existing system.
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#### 4.7 Scalability

The system should support an increasing number of customers, food items, and orders without major performance problems.

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### 5. EXTERNAL INTERFACE REQUIREMENTS

#### 5.1 User Interface

The system should provide the following interfaces:

##### Customer Interface

- Registration page
- Login page
- Home page

- Food menu page
- Food details page
- Cart page
- Checkout page
- Payment page
- Order tracking page
- Order history page
- Feedback page

## Admin Interface

- Admin login
- Dashboard
- Food management
- Category management
- Customer management
- Order management
- Payment details
- Feedback management
- Reports

## Delivery Staff Interface

- Login
- Assigned orders
- Delivery details
- Order status update

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## 5.2 Hardware Interface

The system can be accessed using:

- Desktop computer
  - Laptop
  - Tablet
  - Smartphone
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### 5.3 Software Interface

The system may use:

- Frontend: HTML, CSS, JavaScript
  - Backend: Java / Python / PHP
  - Database: MySQL
  - Server: Apache / suitable application server
  - Browser: Google Chrome / Microsoft Edge / Firefox
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### 5.3 Communication Interface

The system may use an internet connection for communication between the client application, server, database, and payment services.

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## 6. SYSTEM FEATURES

### 6.1 Customer Management

The system manages customer registration, login, profile information, addresses, and order history.

### 6.2 Food Management

The admin can maintain complete information about food items, categories, prices, and availability.

### 6.3 Order Management

The system manages the complete order process from order placement to delivery.

### 6.3 Payment Management

The system records payment details and payment status for customer orders.

### 6.5 Delivery Management

The system provides delivery information and allows delivery staff to update the delivery status.

### 6.4 Feedback Management

Customers can provide feedback and ratings, which can be viewed by the administrator.

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## 7. DATABASE REQUIREMENTS

The system database may contain the following major entities.

### 7.1 User

- User\_ID
- Name
- Email
- Phone
- Address
- Password
- User\_Type

## 7.2 Food

### 7.3

- Food\_ID
- Food\_Name - Category\_ID
- Description
- Price
- Availability
- Image

## 7.3 Category

- Category\_ID
- Category\_Name

## 7.4 Cart

- Cart\_ID
- User\_ID
- Food\_ID
- Quantity

## 7.5 Order

- Order\_ID
- User\_ID
- Order\_Date
- Total\_Amount
- Order\_Status

- Delivery\_Address

#### 7.5 Order\_Item

- Order\_Item\_ID

- Order\_ID

- Food\_ID

- Quantity

- Price

#### 7.7 Payment

- Payment\_ID

- Order\_ID

- Payment\_Method

- Amount

- Payment\_Status

- Payment\_Date

#### 7.6 Delivery

- Delivery\_ID

- Order\_ID

- Delivery\_Staff\_ID

- Delivery\_Status

- Delivery\_Date

#### 7.9 Feedback

- Feedback\_ID

- User\_ID
  - Food\_ID
  - Rating
  - Comments
  - Feedback\_Date
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## 8. SYSTEM WORKFLOW

The basic workflow of the Food Management System is:

Customer Registration/Login



View Food Menu



Select Food Items



Add Items to Cart



Confirm Order



Make Payment



Order Confirmation



Food Preparation



Delivery



Order Completed



Feedback and Rating

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## 9. USE CASE REQUIREMENTS

## Customer Use Cases

- Register
- Login
- View Menu
- Search Food
- Add to Cart
- Place Order
- Make Payment
- Track Order
- View Order History
- Give Feedback

## Admin Use Cases

- Login
- Manage Food
- Manage Categories
- Manage Customers
- Manage Orders
- Manage Payments - Manage Feedback
- Generate Reports

## Delivery Staff Use Cases

- Login
- View Assigned Orders
- View Delivery Details

- Update Delivery Status

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## 10. SYSTEM ARCHITECTURE

The system follows a basic three-layer architecture:

### Presentation Layer

Provides the user interface for customers, administrators, and delivery staff.

### Application Layer

Processes user requests, business logic, order processing, payment processing, and authentication.

### Database Layer

Stores users, food items, orders, payments, delivery details, and feedback.

Architecture Flow:

User Interface

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Application / Business Logic

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Database

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## 11. EXPECTED OUTPUT

The system is expected to provide:

- Easy food menu management
  - Faster order processing
  - Accurate order information
  - Organized customer information
  - Payment record management
  - Delivery status tracking
  - Customer feedback management
  - Reduced manual work
  - Better overall food-service management
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## 12. FUTURE ENHANCEMENTS

The system can be enhanced in the future by adding:

- Mobile application
- Online live order tracking
- AI-based food recommendations
- Personalized menu recommendations
- Multiple restaurant support
- Digital coupons and offers
- Notification through SMS/email

- Advanced sales analytics
  - Inventory and stock management
  - Integration with external payment gateways
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## 13. CONCLUSION

The Food Management System provides an organized and efficient solution for managing food items, customers, orders, payments, deliveries, and feedback.

The system reduces manual activities and provides centralized management of foodrelated information. By implementing the proposed system, customers can conveniently order food while administrators can efficiently manage the overall foodservice process.

The system is designed to be secure, reliable, user-friendly, maintainable, and scalable for future development.

